

Existence of maximal monotone extensions

Proof of Conjecture 00000008847

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Result. Both assertions follow from a general maximal-extension theorem for pairwise conditions on graphs. This theorem permits arbitrary source and target sets. In particular, it covers monotone operators into a continuous dual space, as well as monotone operators on inner-product spaces. No completeness assumption is needed.

1. General graph formulation

Let X, Y be arbitrary sets and let $C(p, q)$ be any predicate on pairs of points $p, q \in X \times Y$. No symmetry, reflexivity, or algebraic condition on C is assumed. Call $S \subseteq X \times Y$ *admissible* if

$$\forall p, q \in S, \quad C(p, q).$$

A set-valued map $A : X \rightarrow \mathcal{P}(Y)$ has graph

$$\text{gra } A = \{(x, u) : u \in A(x)\}.$$

Call A *admissible* when its graph is admissible. A map B extends A when $A(x) \subseteq B(x)$ for every x , equivalently $\text{gra } A \subseteq \text{gra } B$. Empty values and empty graphs are allowed. A map is *maximal admissible* when it is admissible and every admissible extension of it is equal to it.

General theorem. Every admissible map admits a maximal admissible extension. Moreover, a map is maximal admissible if and only if its graph is a maximal element of the family of admissible graphs ordered by inclusion.

2. The chain-union lemma

Let $\mathcal{C}$ be a family of admissible graphs linearly ordered by inclusion. Then $U = \bigcup_{S \in \mathcal{C}} S$ is admissible.

Indeed, choose $p, q \in U$. There are $S, T \in \mathcal{C}$ with $p \in S$ and $q \in T$. Either $S \subseteq T$ or $T \subseteq S$. Thus the two points lie together in one admissible graph, which gives $C(p, q)$. For the empty chain, the union is empty and admissibility is vacuous. This argument uses only that admissibility is a condition on each pair of graph points.

3. Existence and graph-order maximality

Fix an admissible map A and consider

$$\mathcal{M}_A = \{S \subseteq X \times Y : \text{gra } A \subseteq S, S \text{ is admissible}\},$$

ordered by inclusion. This family is nonempty because it contains $\text{gra } A$. A nonempty chain in $\mathcal{M}_A$ has an upper bound in $\mathcal{M}_A$ given by its union: the union contains $\text{gra } A$ and is admissible by Section 2. The empty chain has $\text{gra } A$ as an upper bound. Zorn's lemma therefore gives a maximal element $M \in \mathcal{M}_A$.

Set

$$B(x) = \{u \in Y : (x, u) \in M\}.$$

Then $\text{gra } B = M$, and B is an admissible extension of A . If an admissible map D extends B , its graph contains $\text{gra } A$ and belongs to $\mathcal{M}_A$. Maximality gives $\text{gra } D = M = \text{gra } B$, so $D = B$. This proves the extension assertion.

For the characterization, the constructions

$$A \longmapsto \text{gra } A, \quad S \longmapsto [x \mapsto \{u : (x, u) \in S\}]$$

are inverse bijections between all set-valued maps $X \rightarrow \mathcal{P}(Y)$ and all subsets of $X \times Y$. They preserve inclusion and admissibility. Therefore graph-order maximality is exactly the absence of a proper admissible map extension, in both directions. $\square$

4. Standard monotonicity specializations

Continuous-dual operators. Let E be a real normed space and E^* its continuous dual, the space of continuous real linear maps $E \rightarrow \mathbb{R}$. Take $X = E$, $Y = E^*$, and

$$C((x, u), (y, v)) \iff (u - v)(x - y) \geq 0.$$

The admissible maps are precisely the usual monotone set-valued operators $A : E \rightarrow \mathcal{P}(E^*)$. The general theorem gives a maximal monotone extension of every such operator and the exact graph-order characterization. This applies to every real Banach space; completeness is unnecessary. Evaluation here is actual evaluation of continuous linear functionals, not an assumed surrogate pairing.

Inner-product operators. For a real inner-product space H , take $X = Y = H$ and

$$C((x, u), (y, v)) \iff \langle x - y, u - v \rangle \geq 0.$$

This recovers both conclusions for monotone operators $H \rightarrow \mathcal{P}(H)$, in particular for Hilbert spaces. The general theorem also applies to other source/target pairings by inserting their defining pairwise monotonicity predicate.

5. Scope and formal correspondence

The English and Chinese conjecture both assert existence of maximal extensions and maximality under graph inclusion. The general theorem proves these two assertions for every pairwise graph compatibility condition, with explicit standard monotone-operator instances in Section 4. It does not restrict the original claim to equal source and target spaces or to Hilbert spaces.

No continuity of the set-valued operator, nonempty-domain, full-domain, or completeness assumption is used. Zorn’s lemma uses classical choice. The result asserts existence, not uniqueness or an effective procedure for selecting an extension.

The accompanying Lean 4.19.0 project pins Mathlib to commit

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The principal definitions and results in `lean/Main.lean` are:

- `graph`, `ofGraph`, and both inverse identities for arbitrary source and target types;
- `IsAdmissibleGraph`, the actual condition $\forall p, q \in S, C(p, q)$, and `admissible_iff`, its pointwise map formulation;
- `graph_subset_iff`, the equivalence between valewise map extension and graph inclusion;
- `admissible_sUnion`, the chain-union proof, and `exists_maximal_graph`, Zorn’s lemma with the prescribed graph as seed;
- `maximal_graph_iff`, the precise equivalence with absence of proper admissible map extensions, and `exists_maximal_extension`, the map existence result.

The general final theorem `conjecture_00000008847` combines both conclusions for arbitrary X , Y , and C . No hypothesis asserts an extension or a maximal element. The source then defines the genuine inner-product and continuous-dual compatibility predicates and proves both specialized theorems, `InnerProduct.result` and `Duality.result`. In the dual instance the target is Lean’s type of continuous real linear maps, and `Duality.monotone_iff` identifies the condition with $(u - v)(x - y) \geq 0$.

All three final theorems are kernel checked and audited. There are no additional axioms, incomplete terms, or `native_decide`. The standard logical axioms, including the classical choice used by Zorn’s lemma, are recorded in `VERIFICATION.md`. No auxiliary numerical computation is involved.